A Theory of Computational Coupling Between Intelligent Systems

Toward a General Foundation for Brain-to-Brain Communication

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Status: This is a working draft establishing the core theoretical framework and central definitions. Sections 1, 2, 6, and 8 are intentionally left as structured outlines for further development. Section 3 (Formal Framework) and Section 4 (Falsifiable Predictions) constitute the primary contribution of this draft. Related-work citations are placeholders and require verification before any submission or public circulation.

Abstract

Efforts toward brain-to-brain communication (BBI) and related brain-computer interface (BCI) paradigms have largely proceeded by designing communication *protocols* — fixed encodings, tokenizations, or aligned representational spaces — without first establishing what quantity such a protocol should maximize, or how to measure whether it has succeeded. We argue this mirrors the state of telecommunications engineering prior to Shannon's definition of channel capacity: a field of designed artifacts without a rigorous target. We introduce a general theory of **computational coupling** between intelligent systems — biological or artificial — that defines a substrate-independent, directed, information-theoretic quantity (**coupling capacity**) governing how predictively entangled two systems' internal state trajectories can become, given a bandwidth- and structure-constrained interface between them. We derive three falsifiable predictions from this theory, propose estimators suitable for both simulated multi-agent systems and biological recordings, and outline how brain-to-brain communication, communication protocols, and language itself are best understood as special-case applications of this more general quantity, rather than as the object of study in themselves.

1 Introduction

[Outline — to be expanded]

- **Motivation.** Prior to 1948, telecommunications engineering possessed extensive practical technique but no rigorous definition of *information* or of a channel's fundamental capacity. Shannon's contribution was not a device but a *measurement theory*. We argue BBI and BCI research occupy an analogous position today: substantial engineering effort (tokenization schemes, decoders, aligned embeddings) without a governing quantity that specifies what is being optimized, or what its fundamental limits are.

- **Core reframing.** Communication between two intelligent systems is not the transmission of a preformed message, but the degree to which their internal state trajectories become mutually predictive through a designed or evolved interface. This reframing subsumes, rather than discards, decoding-based and alignment-based approaches as special cases.
- **Central object.** We introduce *coupling capacity*: a directed, dynamic generalization of Shannon channel capacity, defined over trajectories rather than static messages (Section 3).
- **Contribution.** Three falsifiable, cross-domain predictions (Section 4) distinguish this framework from a purely terminological unification of existing subfields.
- **Scope statement.** This paper is a measurement theory, not a system proposal. It does not propose a BBI device, a new tokenization scheme, or a hardware architecture; it proposes what any such system should be evaluated against.

2 Related Work

[Outline — topic areas only; specific citations to be added and verified prior to any public release]

- Information-bottleneck and representation-learning approaches to neural decoding.
- Hyperalignment and cross-subject representational alignment methods.
- Transfer entropy and directed information (Schreiber-style formulations) in time-series analysis.
- Synchronization theory and coupled-oscillator models (Kuramoto-type systems); inter-brain neural coupling ("hyperscanning") during joint attention and natural conversation.
- Active inference and free-energy formulations of communication as mutual prediction-error minimization.
- Emergent communication protocols in multi-agent reinforcement learning.

Positioning: each line of work above formalizes one *instance* of coupling — a fixed substrate, a fixed interface, or a fixed direction — rather than the general quantity governing coupling itself.

3 Formal Framework

3.1 Systems

Let each system $i \in \{1, \ldots, N\}$ (biological or artificial) be characterized by:

- A state space $\mathcal{M}_i$ (a manifold, possibly high-dimensional), with internal state trajectory $x_i(t) \in \mathcal{M}_i$ evolving over a time index t (discrete or continuous).
- A self-predictive model P_i, used by system i to predict its own future state $x_i(t + \Delta)$ from its own past $x_i(\leq t)$.

We deliberately do not assume $\mathcal{M}_i$ is shared, aligned, or even comparable in dimension across systems — a direct departure from hyperalignment-based approaches, which presuppose a common representational geometry.

3.2 Interfaces

An **interface** from system i to system j is a (possibly stochastic) map

$$g_{i\to j} : x_i(\leq t) \;\longmapsto\; s_{i\to j}(t),$$

producing a signal $s_{i\to j}(t)$ that influences the subsequent evolution of x_j. Interfaces are constrained by a bandwidth or structural budget — e.g., a discrete channel with vocabulary size K, a continuous channel with bandwidth B, or a hardware-constrained stimulation protocol. Let $\mathcal{G}$ denote the admissible set of interfaces under a given constraint budget.

3.3 Directed Coupling

We define the directed coupling from i to j, at lag Δ and under interface $g \in \mathcal{G}$, as the transfer entropy (directed information):

$$\mathrm{TE}_{i\to j}(\Delta; g) \;=\; I\big(x_i(t)\,;\, x_j(t+\Delta) \,\big|\, x_j(\leq t)\big), \tag{1}$$

the reduction in uncertainty about j's future state given i's state, beyond what j's own past already provides. This generalizes standard transfer entropy to vector- or manifold-valued states with an explicit lag term, since biological systems exhibit genuine transmission delay and representational drift that a lag-free formulation would misattribute to noise.

3.4 Coupling Capacity

Definition 1 (Coupling Capacity)**.** *The **coupling capacity** from system i to system j is the supremum of directed coupling over admissible interfaces:*

$$C(i \to j) \;=\; \sup_{g\,\in\,\mathcal{G}} \;\mathrm{TE}_{i\to j}(\Delta; g). \tag{2}$$

This is the direct dynamical analogue of Shannon capacity: where Shannon capacity is a supremum over input distributions of mutual information subject to a power or bandwidth constraint, coupling capacity is a supremum over *interface designs* of directed information between two coupled dynamical systems, subject to an analogous constraint.

3.5 Bidirectional Coupling and Asymmetry

Define total coupling as $C_{\text{total}} = C(i \to j) + C(j \to i)$, and an asymmetry index

$$A \;=\; \frac{C(i \to j) - C(j \to i)}{C(i \to j) + C(j \to i)} \;\in\; [-1, 1]. \tag{3}$$

A is predicted to correlate with task-defined role (e.g., "leading" vs. "following," "speaking" vs. "listening"); see Prediction 3 below.

4 Falsifiable Predictions

Prediction 1 (Capacity–Bandwidth Law)**.** *For interfaces parameterized by a scalar bandwidth B, $C(i \to j; B)$ is monotonically increasing and concave in B, saturating at a value set by $\min\big(\dim_{\text{eff}}(\mathcal{M}_i), \dim_{\text{eff}}(\mathcal{M}_j)\big)$ — the smaller of the two systems'* effective *representational dimensionality — rather than by the channel's raw capacity. This is directly falsifiable: if measured capacity continues scaling with channel bandwidth well past the point where either system's effective dimensionality should cap it, the theory as stated is wrong.*

Prediction 2 (Self-Predictive Accuracy Governs Capacity Efficiency). *Let R_i denote system i's self-predictive accuracy — its own model's log-likelihood in predicting its own future state. Capacity achieved per unit bandwidth, $C(i \to j)/B$, is monotonically increasing in R_i and R_j jointly: systems with better internal world models extract more coupling from an identical channel.*

Prediction 3 (Asymmetry Tracks Role). *The asymmetry index A correlates with an externally defined task-role variable across domains (multi-agent reinforcement learning, human dialogue, hyperscanning) — a cross-domain claim, not a redescription of a single-domain result.*

5 Evaluation Metric and Estimation

- **Simulated multi-agent systems.** Full access to ground-truth internal states permits direct estimation via k-nearest-neighbor transfer entropy estimators (Kraskov–Stögbauer–Grassberger–style), or via a *predictive-gain estimator*: train an auxiliary model to predict $x_j(t + \Delta)$ from $x_j(\le t)$ alone, then from $x_j(\le t)$ and $x_i(\le t)$ jointly, and take the log-likelihood gap as the TE estimate. The predictive-gain estimator is expected to be more robust in high dimensions than raw k-NN estimators.
- **Biological recordings (EEG / fMRI hyperscanning).** State trajectories are estimated via dimensionality reduction (PCA or factor analysis on the population signal) prior to applying the same predictive-gain estimator. This setting requires explicit treatment of measurement noise and short-data regimes as a stated limitation, not an afterthought.

6 Falsifiability and Scope

[Outline — to be expanded]

- State explicitly what observation would falsify the framework (Predictions 1–3, Section 4).
- Scope limitation: this framework specifies *how much* two systems can become mutually predictive, not *what* they become predictive about. Content-level questions are downstream and substrate-specific.
- Commitment to Predictions 1–3 as stated prior to running Paper 1, to avoid post-hoc loosening that would render the framework unfalsifiable.

7 Roadmap

[Brief, forward-pointing only — full detail reserved for subsequent papers]

1. **Paper 1:** Validate Predictions 1–2 in a controlled multi-agent reinforcement learning testbed with full ground-truth access.
2. **Paper 2:** Test the same predictions against public human hyperscanning and dialogue datasets.
3. **Paper 3:** Causal manipulation of a theory-predicted variable via non-invasive intervention.
4. **Paper 4 / Prototype:** Use coupling capacity as a training objective to discover a data-derived interface between two systems, culminating in a closed-loop BBI prototype for a narrow task domain.

8 Discussion and Limitations

[Outline — to be expanded: computational cost of estimators at scale; ethical considerations attending causal/stimulation work; relationship to existing capacity-like quantities in neuroscience, e.g. integrated information.]

Draft state: Sections 3 and 4 constitute the completed formal core of this working paper. Sections 1, 2, 6, and 8 are structured outlines pending expansion. No citation in this draft should be treated as verified; all require confirmation against primary sources prior to any external use or submission.